

DHEEPAK S



+91 9043622330
poonamallee, Tiruvallur.



dheep6381@gmail.com
www.linkedin.com/in/dheepak-s

ABOUT ME

I am a final year B.Tech Electronics and Communication Engineering student with a strong passion for Embedded Systems, IoT, Artificial Intelligence, Computer Vision, and Industrial Automation. I enjoy designing intelligent real-time embedded solutions using STM32, ESP32, LoRa, and Edge AI technologies. With experience in research, product development, technical leadership, and innovation, I actively work on solving real-world engineering challenges through interdisciplinary projects, hackathons, and collaborative teamwork.

SKILLS

- Embedded C
- C++
- Python
- STM32
- ESP32 / ESP32-S3
- Embedded Systems
- Edge AI
- Machine Learning
- OpenCV
- Computer Vision
- LoRa Communication
- IoT System Design
- Sensor Interfacing
- CAN Protocol
- Digital Electronics
- Industrial Automation
- MATLAB
- HTML, CSS & JavaScript
- Proteus

EXPERIENCE

- Internship at Hyundai pvt. Ltd. MAY-2025
Internship at Kyungshin Industrial Motherson Limited (KIML).
 - Embedded Systems & AI Research Developer (2025–Present)
 - Developed AI-enabled embedded systems using STM32 and ESP32 platforms.
 - Designed intelligent IoT applications integrating Edge AI, LoRa communication, and Machine Learning.
 - Worked on embedded hardware, sensor interfacing, and real-time automation projects.
 - Collaborated with multidisciplinary teams in research, hackathons, and technical innovation.

PROJECTS AND HANDS ON

- PyroSentinal-Early Forest Fire Detection MAY-2026
- Edge AI Driver Emotion Recognition & Vehicle Control MAY-2026
- Interactive smith chart JAN-2026
- Wearable Non-Invasive Glucose Monitoring System NOV-2024
- Robotic Arm using STM-32F103RB NOV-2025
- IoT Attendance Monitoring System using ESP-32 S3 NOV-2024

HACKATHONS

- AICTE IIC Regional Meet 2025 – Shortlisted for YUKTI & APF Grant Competition (Round 2) with the Non-Invasive Musculoskeletal Diagnostic System. NOV-2025
- IASF EduAIthon 2026 – GITAM University. JAN-2026
- ELECTROTHON 2026 – Rathinam Technical Campus. MAR-2026
- Ideathon 2026 – PSR Engineering College. APR-2026
- Nanochip Hackathon 2025 – Developed a Wearable Non-Invasive Glucose Monitoring System. APR-2025
- VINTRAA Technical Challenge 2025 – Participant. OCT-2025

PATENTS

PyroSentinal

AI-Powered LoRa-Based Low Power Forest Fire Early Detection and Intelligent Communication System Using PSO-Optimized Mesh Architecture 2025-2026

SMITH CHART.IO

INTERACTIVE WEB-BASED SMITH CHART ANALYZER WITH INTEGRATED OFFLINE LLM ASSISTANCE FOR REAL-TIME RF IMPEDANCE MATCHING

PUBLICATIONS

An Adaptive Radiomics-Guided Spatial Clustering Framework for Lung Tumor Segmentation from CT Images. 2025-2026

EDUCATION & CERTIFICATIONS

Bachelor of Technology

Electronics and communication Engineering
Kalasalingam Academy of research and education
CGPA : 9.02

Minors in unmanned Ariel Vehicles(UAV)

Department of Aeronautical Engineering
Kalasalingam Academy of research and education

ACTIVE PARTICIPATIONS

- Public Relations Team Lead – ACM SIGBED Student Chapter
- IEEE Published Author
- Indian Patent Holder
- AICTE IIC Regional Meet (YUKTI & APF Grant – Round 2)

EXTRA CURRICULAR ACTIVITIES

- Participating hackathons
- Independent projects

SOFT SKILLS

- Problem Solving
- Analytical Thinking
- Leadership
- Team Collaboration
- Communication
- Adaptability
- Critical Thinking
- Self-Learning